ModelIt!

Student Activity Pack

Restoring Ecosystems — Data-Driven Recovery

Using Science to Heal Damaged Landscapes

Grade NE

NGSS HS-LS2-6, HS-LS2-7, HS-ESS3-3, HS-ESS3-4, HS-ETS1-4

Stability and Change

STUDENT NAME _____ DATE _____

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **In Level 1, you modeled how removing a keystone species collapses an ecosystem. Now think about the reverse: what does it take to REBUILD one?**

2. **Why do you think some ecosystem restoration projects succeed and others fail?**

3. **What data would you need to determine whether a damaged ecosystem is recovering?**

4. **What do you think 'self-sustaining' means in the context of an ecosystem — when can scientists stop actively managing it?**

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Restoration Investment | Keystone Species Reintroduced | Human Development Pressure | Habitat Quality | Native Species Diversity | Invasive Species Pressure | Water Quality | Recovery Rate | Self-Sustaining Threshold

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Full Investment

Maximum resources, keystone reintroduction, minimal pressure

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Underfunded Restoration

Low funding, some keystone species, no development pressure

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Development Pressure

Full investment against active habitat destruction

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Climate Variability — Year-to-year changes in temperature and precipitation patterns that can accelerate or slow recovery

Community Engagement — Level of local public support and volunteer involvement in restoration — affects long-term protection

Soil Microbiome Health — The diversity and activity of soil fungi, bacteria, and invertebrates that drive nutrient cycling and plant establishment

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

How do scientists know when a damaged ecosystem has healed enough to sustain itself — and what happens if we stop investing too early? Use evidence from your simulation to support your answers.

1. **How does the reinforcing loop between Habitat Quality and Native Species Diversity create a tipping point?**

2. **Why is the Early Withdrawal scenario the most important one for real-world conservation planners?**

3. **Compare the Full Investment and Underfunded scenarios — what does this tell you about the relationship between money and recovery speed?**

4. **How does keystone species reintroduction act as a 'force multiplier' in restoration?**

5. **If you were advising a state conservation board, what single metric would you track to decide when it is safe to reduce funding — and why?**

8 STEM Challenge: Restoration Proposal Pitch

ModelIt!

YOUR ENGINEERING MISSION

Select a real degraded ecosystem in your state, research its current condition, and design a data-driven restoration plan that you pitch to a panel of judges acting as a state conservation board.

THE SCENARIO

Your state's Department of Natural Resources has a limited budget and must choose which degraded ecosystem to restore first. Your team will research a real site, use model data to design a restoration plan with a timeline and budget, and pitch it to a panel. The most compelling, evidence-based pitch wins funding.

Engineering Design Thinking

1. WHAT DATA SOURCES DID YOU USE TO ASSESS THE CURRENT CONDITION OF YOUR ECOSYSTEM?

2. HOW WILL YOU ALLOCATE YOUR RESTORATION BUDGET ACROSS DIFFERENT INTERVENTIONS (PLANTING, SPECIES REINTRODUCTION, INVASIVE REMOVAL, MONITORING)?

3. WHAT METRICS WILL YOU TRACK TO KNOW WHEN THE SELF-SUSTAINING THRESHOLD HAS BEEN REACHED?

4. HOW DOES YOUR PLAN ACCOUNT FOR ONGOING HUMAN DEVELOPMENT PRESSURE NEAR THE SITE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Restoration Ecologists design and manage projects that return degraded ecosystems to healthy function. They work with government agencies, conservation nonprofits, and engineering firms to plan reintroductions, monitor recovery, and manage invasive species. They earn $60,000–$100,000/year.